

Phase-Field Crystal Framework for Single-Crystal Copper: Summary

Project Memo

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1 Background and Objectives

The project replicates the ductile-fracture framework of Xu *et al.*[1] to predict the tensile response of single-crystal copper (Cu) informed by EBSD observations. The primary objectives are: (i) build synthetic or EBSD-derived [111] Cu structures with defects, (ii) couple Phase-Field Crystal (PFC) density evolution with phase-field fracture and crystal plasticity, (iii) solve mechanical equilibrium for anisotropic FCC copper, evolve plasticity and cracks, and export results for Ovito/ParaView, and (iv) compare simulated crack paths, stress-strain curves, and fatigue metrics (Paris, Coffin-Manson) with experiments.

2 Energy Functional and Constraints

State variables include the PFC density ψ , crack field $\phi \in [0, 1]$, plastic strain ε^p , and grain/orientation markers η_g . The free energy reads

$$F = \int_{\Omega} [(1 - \phi)^2 f_{\text{elas}}^+ + f_{\text{elas}}^- + f_{\text{frac}} + f_{\text{grain}} + f_{\text{PFC}}] d\Omega. \quad (1)$$

Elastic energy employs the cubic stiffness tensor C_{ijkl} , rotated to the local orientation R via $C'_{ijkl} = R_{ip}R_{jq}R_{kr}R_{ls}C_{pqrs}$. Spectral decomposition of the strain tensor provides tensile/compressive splits: $\varepsilon = Q\Lambda Q^\top$, and Λ^+ keeps positive eigenvalues. Here ψ is the dimensionless PFC density deviation, ϕ is the damage/crack variable, ε^p is the plastic strain, η_g labels grain orientation, $f_{\text{elas}}^\pm$ are the positive/negative parts of the elastic energy density built from $\varepsilon - \varepsilon^p$, and f_{PFC} captures lattice-scale ordering.

Fracture energy

$$f_{\text{frac}} = G_c(\eta_g, \varepsilon_{\text{eq}}) \left[\frac{\phi^2}{2\ell_0} + \frac{\ell_0}{2} |\nabla \phi|^2 \right] \quad (2)$$

uses a toughness degradation law

$$G_c = G_{c0} \left(0.5 - g_{\text{res}} \tanh \left[2 \left(\frac{\varepsilon_{\text{eq}}}{\varepsilon_{1/2}} - 1 \right) \right] + 0.5 + g_{\text{res}} \right), \quad (3)$$

which lowers G_c with accumulated plastic strain. Parameters: G_{c0} initial toughness, ℓ_0 regularization length, g_{res} residual toughness ratio, and $\varepsilon_{1/2}$ the equivalent plastic strain at half-toughness. The equivalent plastic strain ε_{eq} is derived from the PFC density gradients or an explicit plastic field, matching the code path in **PFCCoupling**. Crack driving compares tensile elastic energy density with the critical fracture energy density G_c/ℓ_0 , so the numerical driving term uses $H\ell_0/(G_c + \epsilon)$; this reintroduces the characteristic length to maintain a dimensionless ratio and avoid blow-up.

Plastic surrogates and anisotropic driving Mechanical and PFC contributions are blended, with mechanical part driven by FCC slip RSS (orthogonal $\{111\}\langle 110 \rangle$, sign-preserving, with yield threshold):

$$\hat{\tau}_{\text{RSS}} = \max_{m,n \in \{111\}\langle 110 \rangle} |m \cdot (\varepsilon n)|, \quad \hat{g} = \frac{\sqrt{\sum_i (\partial_i \psi)^2}}{\max |\cdot|}, \quad (4)$$

$$\varepsilon_{\text{eq}} = w_{\text{mech}} \frac{\hat{\tau}_{\text{RSS}}}{\max \hat{\tau}_{\text{RSS}}} + (1 - w_{\text{mech}}) \hat{g}, \quad (5)$$

$$\varepsilon_{\text{flow}}^p = \text{flow}(\hat{\tau}_{\text{RSS}} - \tau_y) \text{sign}(m \cdot \varepsilon n) \text{sym}(m \otimes n), \quad (6)$$

$$\varepsilon_{\text{eq}}^{\text{acc}} \leftarrow \varepsilon_{\text{eq}}^{\text{acc}} + \Delta t \gamma \varepsilon_{\text{eq}}, \quad \varepsilon^p \leftarrow \varepsilon^p + \Delta t \gamma \varepsilon_{\text{flow}}^p, \quad (7)$$

$$\text{flow}(x) = \max\left(0, \frac{x}{1 - \tau_y}\right) s_{\text{flow}}, \quad (8)$$

$$\varepsilon_{\text{dir},i} = w_{\text{mech}} [\varepsilon_{ii}]_+ + (1 - w_{\text{mech}}) \frac{|\partial_i \psi|}{\max_j |\partial_j \psi|}, \quad (9)$$

where $w_{\text{mech}} \in [0, 1]$ (default 0.9) and $[\cdot]_+$ clamps to tension. The loading-axis component $\varepsilon_{\text{dir},L}$ boosts both the history and crack driving:

$$H \leftarrow \max(H, f_{\text{elas}}^+ (1 + k_{\text{dir}} \varepsilon_{\text{dir},L})), \quad \mathcal{D} = \frac{H \ell_0}{G_c + \epsilon} (1 - \phi) (1 + k_{\text{dir}} \varepsilon_{\text{dir},L}), \quad (10)$$

with k_{dir} tuning anisotropy. When stress coupling is enabled, the chemical potential gets an additive term

$$\mu_{\text{extra}} = \alpha \frac{\sigma_{\text{vm}}}{\max |\sigma_{\text{vm}}|}, \quad (11)$$

so high von Mises regions accelerate PFC evolution. In mechanics, the accumulated plastic tensor ε^p enters as an eigenstrain, so the stress is $C : (\varepsilon - \varepsilon^p)$; the accumulated scalar $\varepsilon_{\text{eq}}^{\text{acc}}$ degrades toughness $G_c(\varepsilon_{\text{eq}}^{\text{acc}})$ (optionally weakened on grain boundaries).

The PFC energy follows

$$f_{\text{PFC}} = \frac{\psi}{2} [r + (q_0^2 + \nabla^2)^2] \psi + \frac{u}{4} \psi^4, \quad (12)$$

with constraints (density conservation, grain fractions, plastic incompressibility) enforced through Lagrange multipliers in the modified TDGL formulation. Here r is the undercooling parameter that sets the ordering tendency, q_0 is the principal reciprocal lattice magnitude, and u penalizes large density excursions; the noise amplitude only seeds initial defects and does not enter the energy.

3 Mechanical Equilibrium

Small-strain equilibrium is imposed:

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}, \quad \boldsymbol{\sigma} = C'(\mathbf{x}) : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p), \quad (13)$$

with periodic in-plane boundaries and prescribed macroscopic strain $\boldsymbol{\varepsilon}^{\text{macro}}$ in the loading direction. A conjugate-gradient linear operator solves for displacement $\mathbf{u}$, yielding total strain $\boldsymbol{\varepsilon} = \nabla_s \mathbf{u} + \boldsymbol{\varepsilon}^{\text{macro}}$. The rotated tensor $C'(\mathbf{x})$ comes from local orientation $R(\mathbf{x})$; $\boldsymbol{\varepsilon}^p$ is either driven by PFC gradients or provided explicitly.

4 Phase-Field Crystal Dynamics

PFC density obeys conserved dynamics:

$$\frac{\partial \psi}{\partial t} = -\nabla^2 \mu, \quad \mu = r\psi + (q_0^2 + \nabla^2)^2 \psi + u\psi^3, \quad (14)$$

implemented spectrally with FFTs to evaluate $(q_0^2 - k^2)^2$ and advance ψ semi-implicitly.

5 Crack Evolution

A history variable $H(\mathbf{x}) = \max_{t' \leq t} f_{\text{elas}}^+(\mathbf{x}, t')$ accumulates tensile energy. Crack driving force combines energy and toughness:

$$\mathcal{D} = \frac{H \ell_0}{G_c + \epsilon} (1 - \phi), \quad (15)$$

which compares the current tensile energy density to the critical fracture energy density G_c/ℓ_0 (here $\ell_0 \approx 3 \times 10^{-6}$ m). The evolution law $\dot{\phi} = L_\phi \mathcal{D}$ is clamped to $[0, 1]$, so cracks grow faster where energy is high and toughness is low. Here H stores the maximum tensile elastic energy density seen so far, ϵ is a small numerical residual stiffness, and L_ϕ is the kinetic coefficient for the crack field.

6 Numerical Workflow

1. **Structure generation:** ‘Cu111StructureBuilder’ constructs grids ($32 \times 32 \times 16$, spacing 5×10^{-7} m) with [111] orientation and random defects for testing; EBSD data can replace the synthetic mask in future runs. Defect seeding uses weighted/random seeds (and optional line defects) projected via anisotropic Gaussians to ψ /crack/plastic; optional notch boxes pre-set high crack values as starters.
2. **Alternating solver:** The cyclic load path is triangular $0 \rightarrow +\varepsilon_{\text{max}} \rightarrow 0 \rightarrow -\varepsilon_{\text{max}} \rightarrow 0$, each segment subdivided into N_{seg} steps. At every increment: solve mechanical equilibrium (CG); blend mechanical/PFC plastic surrogates and relax plastic scalar + directional vector; update crack with anisotropic boost; advance PFC with optional stress-coupled μ_{extra} .

3. **Macro strain with Poisson contraction:** Applied strain is $(\varepsilon, -\nu\varepsilon, -\nu\varepsilon)$ (default $\nu = 0.34$ for Cu); the same macro strain is used in PFC k -update, VTK warp, and LAMMPS dump.
4. **Toughness scaling:** A scalar s_{Gc} rescales g_c and g_{res} (e.g., $s < 1$ to promote cracking), leaving $\ell_0, k, \epsilon_{1/2}$ unchanged.
5. **Outputs:** Every few steps dump VTK frames (deformed coordinates) and after each cycle dump LAMMPS and optional ‘data’ snapshots. Scalar/vector fields include crack, plastic (scalar + directional), ψ , stress (tensor and von Mises), displacement, and masks (defect/line/notch). CSV logs store per-cycle energy, mean crack/plastic, crack increment, plastic range, and (optionally) a stepwise macro stress–strain curve $\{\varepsilon, \bar{\sigma}_{xx}, \bar{\sigma}_{yy}, \bar{\sigma}_{zz}, \bar{\sigma}_{\text{vm}}\}$; runs stop early if $\langle \phi \rangle$ exceeds a failure threshold.
6. **Fatigue metrics:** Fit crack increment per cycle vs cycle count (Paris-like slope) and plastic strain range vs $2N$ (Coffin–Manson slope) from the logged series.

7 Defect Seeding and Initialization (New)

Weighted random and deterministic line seeds A defect configuration specifies: density ρ_d , region bounds Ω_d , type probabilities p_k (vacancy/interstitial/dislocation), optional weight field $w(\mathbf{x})$, and optional line segments $\{\Gamma_\ell\}$. The total target seeds is $N = \min(\rho_d|\Omega_d|, N_{\text{max}})$. We first discretize each line Γ_ℓ into points $\mathbf{x}_i$ with spacing Δs and mark them as dislocation seeds (anisotropic kernel along the line); the remaining N_{rem} seeds are drawn from a categorical distribution over types and sampled positions:

$$P(\mathbf{x}) = \frac{w(\mathbf{x}) \chi_{\Omega_d}(\mathbf{x})}{\int_{\Omega_d} w(\mathbf{x}) d\mathbf{x}}, \quad w(\mathbf{x}) = \begin{cases} 0, & w < w_{\text{th}} \\ w, & \text{else} \end{cases}, \quad (16)$$

falling back to uniform if w is absent. Directions are random or sampled from FCC slip systems $\langle 110 \rangle$.

Gaussian projection to fields Each seed contributes a 3D anisotropic Gaussian kernel to ψ , crack, and plastic fields. For a seed at $\mathbf{x}_0$ with kernel widths $\boldsymbol{\sigma}$ and weight A :

$$g(\mathbf{x}) = A \exp \left[- \sum_{i=1}^3 \frac{(x_i - x_{0,i})^2}{2\sigma_i^2} \right], \quad \psi \leftarrow \psi + g, \quad \phi \leftarrow \phi + A_\phi g, \quad \varepsilon^p \leftarrow \varepsilon^p + A_p g. \quad (17)$$

Vacancy uses negative A , interstitial/dislocation use positive; line seeds use elongated $\boldsymbol{\sigma}$ along the line direction. After accumulation, ψ is rescaled to a target mean/peak ($\bar{\psi} \rightarrow \bar{\psi}_{\text{tgt}}$, $\max|\psi| \rightarrow \psi_{\text{peak}}$), and fields are clipped to $[0, 1]$. Masks `defect_mask` (all seeds) and `line_mask` (line seeds) are written for visualization.

EBSD orientation ingestion The builder accepts an external orientation map $R(\mathbf{x})$ (e.g., EBSD) to replace the synthetic [111] field, ensuring stiffness rotation and subsequent mechanics follow the measured texture.

Per-step VTK sequence A smoke test outputs VTK per time step with deformed coordinates (`displacement_total`) to allow ParaView time-series playback; final snapshots include both deformed and undeformed grids for easy comparison.

8 Outlook

The framework replicates Fig. 1–4 behavior from Xu *et al.*[1] at the single-crystal scale. Next steps include (i) ingesting actual EBSD orientation/defect maps, (ii) swapping the CG equilibrium for FFT-based Lippmann–Schwinger or FEM for faster convergence and non-periodic BCs, and (iii) embedding full crystal-plasticity slip kinetics and validated fracture history laws.

参考文献

- [1] F. Xue *et al.*, “Phase-field framework with constraints and its applications to ductile fracture in polycrystals and fatigue,” *npj Computational Materials* 8, 18 (2022).

9 Recent Code Updates (Summary)

- 塑性与方向场：新增 `plastic_measures` 计算 ε_{eq} 与方向分量，机械 von Mises/轴向应变与 PFC 梯度按权重混合，默认机械占比 0.9。
- 方向性裂纹驱动：加载轴塑性正分量放大历史能量和驱动力 (`dir_coupling=0.8`)，抑制非加载方向裂纹。
- 应力耦合 PFC：von Mises 应力归一化后作为 μ_{extra} 加入化学势，提升高应力区的晶格演化响应。
- 求解与载荷：采用对称三角循环载荷 ($0 \rightarrow +\varepsilon_{max} \rightarrow 0 \rightarrow -\varepsilon_{max} \rightarrow 0$)，可设置失败阈值提前终止；跟踪每周期塑性范围 $\Delta\varepsilon_p$ 和裂纹增量。
- VTK/LAMMPS 输出扩展：输出塑性方向分量、应力张量/ von Mises，附带归一化场 (`crack_norm`, `plastic_vec_norm`, `stress_vm_norm`, `disp_total_norm` 等) 便于统一色标；VTK 现默认二进制 STRUCTURED_GRID (变形坐标)，避免 ASCII 读入报错，LAMMPS 去重了多余的 `plastic_vec` 列。
- 结构缺陷默认幅值调至 0.12，PFC 截断默认 $|\psi| \leq 1.2$ ，方便在高载下保持稳定。

10 Practical Simulation Recipe

1. **结构初始化**: 用 `Cu111StructureBuilder` 构造网格 (默认 [111] 单晶, 或提供 grain labels/orientation map)。缺陷幅值默认 0.12, 噪声注入 ψ ; 可直接替换为 EBSD 导出的取向场。
2. **求解循环**: `run_virtual_cycles` 采用三角载荷 $0 \rightarrow +\varepsilon_{\max} \rightarrow 0 \rightarrow -\varepsilon_{\max} \rightarrow 0$, 默认每段 50 步, 可多周期。每步执行: 力学求解 (CG); 塑性标量 + 方向分量松弛; 裂纹更新 (方向增益); PFC 更新 (含应力耦合 μ_{extra})。
3. **参数提示**: `mech_plastic_weight=0.9, dir_coupling=0.8, plastic_relax=0.2, clip=1.2, failure_threshold=0.98`; 必要时调节应力耦合系数 (`mu_extra_from_stress`) 或加载轴 `load_axis`。
4. **输出与监控**: 每 5 步 VTK 帧 (`anim_frame*.vtk`) 和每周期 LAMMPS dump (`virtual_cycle*.lammprj`) CSV 日志记录能量、裂纹均值/增量、塑性均值/范围, 裂纹均值超阈值将提前停止。

11 Field/Output Guide

- **VTK frame files** (`sim/tests/virtual_cycle_vtk/anim_frame*.vtk`): 二进制 STRUCTURED_GRID (若启用变形坐标), 含下列标量/向量; 加载时关闭 per-timestep rescale, 固定色标 (裂纹 0-0.3 或 0-1, 塑性/应力 0-1)。
- ψ : PFC 密度偏差, 红/蓝表示高/低密度或缺陷核; 受应力耦合影响会在高应力区演化。
- `crack, crack_norm, crack_clamp03`: 裂纹变量 (0-1), `clamp03` 便于观察萌生阶段, `norm` 便于固定色标。
- `plastic` (标量), `plastic_vec` (方向): 来自机械应变 + PFC 梯度; `plastic_vec_x_norm` 可直接用于加载方向色标, `plastic_vec_mag` 为模值。
- `stress_vm, stress_vm_norm`: von Mises 应力, 可用于判断高应力区; 亦用于 μ_{extra} 。
- `displacement, displacement_total`: 微观/总位移; `disp_total_norm` 提供归一化幅值, 避免色标跳变。
- **LAMMPS dump** (`sim/tests/virtual_cycle_0001.lammprj`): 列顺序为 `id, type, x, y, z, crack, plastic, psi, plastic_xx, plastic_yy, plastic_zz, stress_vm, stress_xx, stress_yy, stress_zz` (若场缺失则省略), 坐标包含宏 + 微观位移。
- **CSV 日志** (`sim/tests/virtual_cycle.csv`): `cycle` (循环编号)、`energy` (总能量)、`crack_mean` (裂纹均值)、`plastic_mean` (塑性均值)、`crack_delta` (该周期裂纹增量)、`plastic_range` (该周期塑性范围 $\Delta\varepsilon_p$)。

- 归一化场均在 VTK 中提供，便于在 ParaView 固定色标（range 0–1 或 0–0.3）播放时间序列。

12 实验流程与数据对应

- **载荷与网格**: 默认单晶网格 $32 \times 32 \times 16$ ，缺陷幅值 0.12；三角循环载荷 ($0 \rightarrow +\epsilon_{\max} \rightarrow 0 \rightarrow -\epsilon_{\max} \rightarrow 0$)，每段 50 步，`max_strain=0.08`，可设置 `failure_threshold` (0.98) 提前终止。
- **周期统计**: `energy` 为周期末总能量；`crack_mean` 代表损伤程度(接近 1 表示贯穿)；`crack_delta` 是相邻周期裂纹增量，用于 Paris 拟合；`plastic_mean` 为塑性均值；`plastic_range` 为该周期塑性最大-最小 ($\Delta\epsilon_p$ 替代量，用于 Coffin–Manson 拟合)。
- **可视化要点**: 在 ParaView/Ovito 打开 VTK 或 LAMMPS dump 时，优先查看 `crack_clamp03` (萌生)、`plastic_vec_x_norm`(加载方向塑性)、`stress_vm_norm`(等效应力)、`disp_total_norm` (总位移)。固定色标 0–1 便于跨帧比较。
- **Coffin–Manson/Paris 指标**: 拟合对数斜率 $d \log(\Delta\epsilon_p)/d \log(2N)$ 得到 Coffin–Manson；拟合 $\log(\Delta\phi)/\log(N)$ 得到 Paris-like 系数，均由 `sim/tests/virtual_cycle.py` 自动计算并打印。

13 当前结果快照（截至 2026-02-09）

- **D1 全量门禁通过**: `sim/tests/regress_runs/2026-02-09/d1_full_gate/summary.json`，总状态 `passed=true`。验收矩阵为 `phase2_full=true, multi_align_full=5/5, seed_robustness=6/6`。
- **D2 裂纹局部化 + 能量密度门禁通过**: `sim/tests/regress_runs/2026-02-09/d2_localization_energy/summary.json`，`cycles_completed=6, crack_delta_total=2.396875, crack_localization_index_peak=5.1072`，并输出 6 个能量相关 VTK 字段。
- **D3 多物理矩阵通过**: `sim/tests/regress_runs/2026-02-09/d3_multiphysics_matrix_smoke/summary.json`，`case_total=3, passed_count=3`；包含 2 个正向裂纹触发 case 与 1 个负对照抑制 case。
- **Release D2+D3 quick+seedfull 通过**: `sim/tests/regress_runs/2026-02-09/release_d2_d3_quick_seedfull/summary.json`，`acceptance.d2_localization=true, acceptance.d3_matrix=true`，并且两批 seed (41–46) 均 `all_seed_gate_passed=true`。

14 全量基准测试矩阵（脚本级）

- `sim/tests/regress_microstrain.py`: 线弹性微应变比值回归；最新通过产物 `sim/tests/regress_runs/2026-02-09/microstrain/summary.json` (`stress_ratio  $\approx$  2.0`)。

- `sim/tests/regress_gnd.py`: GND/Nye 线性响应回归;最新通过产物 `sim/tests/regress_runs/2026-02-08/gnd_growth` (双梯度比值约 2)。
- `sim/tests/regress_gnd_cycle.py`: 低幅循环 GND 增长回归;最新通过产物 `sim/tests/regress_runs/2026-02-08/gnd_growth` (`gnd_growth>0`、`accum_growth>0`)。
- `sim/tests/regress_bc_crack.py`: 边界/裂纹小规模回归;最新通过产物 `sim/tests/regress_runs/2026-02-08/bc_crack`。
- `sim/tests/regress_bc_crack_large.py`: 大网格边界/裂纹回归;最新通过产物 `sim/tests/regress_runs/2026-02-08/bc_crack_large`。
- `sim/tests/regress_bc_crack_micron.py`: 微米尺度缩放回归;最新通过产物 `sim/tests/regress_runs/2026-02-08/bc_crack_micron`。
- `sim/tests/regress_all.py`: 边界裂纹三件套统一入口;最新通过产物 `sim/tests/regress_runs/2026-02-08/all` (`small/large/micron` 全通过)。
- `sim/tests/run_phase1_suite.py`: Phase-1 一键回归;最新通过产物 `sim/tests/regress_runs/2026-02-08/phase1_suite` (`fresh venv` 复现通过)。
- `sim/tests/regress_phase2.py`: Phase-2 综合门禁编排;最新通过产物 `sim/tests/regress_runs/2026-02-08/phase2` (`exp + D2` 同时开启)。
- `sim/tests/regress_exp_alignment.py`: 单工况实验对齐回归;最新通过产物 `sim/tests/regress_runs/2026-02-08/exp_alignment` (`rmse_tau=28.45 MPa`, `rmse_gamma=0.003889`)。
- `sim/tests/regress_energy_consistency.py`: 能量/裂纹/塑性一致性门禁; 最新通过产物 `sim/tests/regress_runs/2026-02-08/energy_gate_smoke/summary.json` (无能量下降和反转计数)。
- `sim/tests/regress_exp_alignment_multi.py`: 多工况实验对齐回归;最新通过产物 `sim/tests/regress_runs/2026-02-08/exp_alignment_multi` (5/5 条件通过)。
- `sim/tests/repeat_crack_onset_seeds.py`: seed 稳健性批次回归;最新通过产物 `sim/tests/regress_runs/2026-02-09/d1_full_gate/seed_batch_2/summary.json` (两批均 3/3)。
- `sim/tests/regress_d2_localization_energy.py`: D2 裂纹局部化触发与能量场输出回归; 最新通过产物 `sim/tests/regress_runs/2026-02-09/d2_localization_energy/summary.json`。
- `sim/tests/regress_d3_multiphysics_matrix.py`: D3 正/负多物理矩阵回归; 最新通过产物 `sim/tests/regress_runs/2026-02-09/d3_multiphysics_matrix_smoke/summary.json` (3/3 通过)。
- `sim/tests/run_d1_full_gate.py`: D1 全量编排入口;最新通过产物 `sim/tests/regress_runs/2026-02-09/d1_full_gate` (验收矩阵全绿)。

- `sim/tests/run_release_baseline_week4.py`: release 编排 (Phase-2 + D2 + D3 + seed); 最新通过产物 `sim/tests/regress_runs/2026-02-09/release_d2_d3_quick_seedfull/bundle_summary.js`
- `sim/tests/run_ci_smoke.py`: CI 最小门禁链路; 最新通过产物 `sim/tests/regress_runs/2026-02-08/ci_` (3 个子任务全部通过)。
- `sim/tests/run_seed_robustness_20.py`: 20-seed 分批运行模板; 最新通过产物 `sim/tests/regress_runs/` (smoke 模式通过)。
- `sim/tests/summarize_seed_robustness_ci.py`: seed CI 统计聚合; 最新通过产物 `sim/tests/regress_run` (`all_seed_gate_passed=true`)。

15 尚需完善与下一步

- **D2/D3 全量化**: 当前已有 quick+seedfull 基线, 下一步应固定 full profile 常态门禁, 并保证与 quick 指标一致。
- **Seed 统计扩容**: 将 seed 稳健性从 6 seeds 扩展到 20 seeds full-case, 产出稳定的 Wilson 区间与聚合报告。
- **D3 负对照增强**: 增加更多“无缺口/弱驱动/不同晶向”负样本, 降低误触发风险并提升外推稳健性。
- **长周期验证**: 建立 > 10 cycles 的 D2+D3 联合回归, 补齐能量密度与裂纹局部化随周期演化证据链。
- **发布摘要自动化**: 在现有 `bundle_summary.json` 基础上生成统一 markdown/tex 摘要, 减少手工整理成本。